

MTH:2310 DISCRETE MATH EXAM 3 REVIEW

- **Recommended Preparation**

- Re-work examples from class.
- Work/re-work suggested problems from the book.
- Re-work quiz problems. Understand and learn from mistakes.
- Go over this study guide and make sure you know and understand every point on this study guide.
- Practice showing all your steps. Make solutions organized and be explicit/intentional in your writing.
- Replicate the exam environment. Can you do relevant problems with little to no help?

2.4 Sequences and Summations

- Be able to find the first N terms of a sequence.
- Be able to completely describe a sequence using words, or as an equation $a_n = f(n)$.
- Be able to show that a sequence is a solution of a given recurrence relation.
- Be able to evaluate single sums and double sums.
- Be able to use provided closed forms to evaluate special sums.

Ex. Section 2.4, Exercises 4, 10, 13, 15, 29, 31, 34

5.3 Recursive Definitions

- Be able to interpret a recursively defined objects like functions, sets, and sequences.
- Be able to give a complete recursive definition for a function, set, or sequence.

Ex. Section 5.3, Exercises 3, 4, 7, 8, 23, 24

8.2 Solving Linear Recurrence Relations

- Be able to identify the degree of a recurrence relation.
- Be able to determine whether a recurrence relation is linear and/or homogeneous and/or has constant coefficients.

- Be able to solve any degree 2 linear homogeneous recurrence relation with constant coefficients.
- Be able to find the general form of the solution to a degree k linear homogeneous recurrence relation with constant coefficients.

Ex. Section 8.2, Exercises 1 - 4*, 12, 13, 21*, 22*

Problems marked with a () were not on the original Suggested Problems list. Problems 2 and 4 are very similar to 1 and 3. Problems 21 and 22 give more practice with general form for solution to degree k recurrence relations when the roots are known.*

5.1 Mathematical Induction

- Be able construct a proof by (weak) mathematical induction following the template found on page 335.
- **The format of your proof is almost as important as the content.**

Ex. Section 5.1, Exercises 5 - 7, 21, 22.

Bonus. Prove that for every positive integer n , $1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$.

Bonus. Prove that for every positive integer n , $2^1 + 2^2 + \dots + 2^n = 2^{n+1} - 2$.

Bonus. Prove that for every non-negative integer n , 9 divides $4^{3n} + 8$.

4.1 Basic Modular Arithmetic

- Be able to simplify expressions using the modulo operator.
- Know when two integers are congruent modulo some positive integer m .
- Be able to find the multiplicative inverse of an integer modulo some positive integer m if it exists.

Ex. Section 4.1, Exercises 17, 26, 30 - 33, 37 - 39

Bonus. Determine whether 4^{-1} exists modulo 7. Does it exist modulo 8? Justify your answer.

4.3 Primes and Greatest Common Divisors

- Be able to use Trial Division to determine whether a given integer is prime or composite.
- Be able to find the prime factorization of a given integer.
- Be able to find the greatest common divisor of two integers using their prime factorization.
- Be able to find the greatest common divisor of two integers using the Euclidean algorithm.

Ex. Section 4.3, Exercises 2 - 5, 24, 32, 33

4.4 Solving Linear Congruences

- Know when a system of linear congruences has a unique solution.
- Be able to use the Chinese Remainder Theorem to solve a system of linear congruences.
- Be able to use the substitution method to solve a system of linear congruences.

Ex. Section 4.4, Exercises 5, 6, 20 - 24